

E.2 Quantum physics [HL only]

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **25 marks**

$h = 6.63 \times 10^{-34} \text{ J s}$; $m_e = 9.11 \times 10^{-31} \text{ kg}$; $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$. Show your working.

Question 1 — The doorbell that refused (10 marks)

A light-activated doorbell uses a metal photocell with work function $\Phi = 2.2 \text{ eV}$. A visitor shines an extremely bright red laser pointer (photon energy 1.9 eV) at it. Nothing happens. Her child's feeble ultraviolet keyring torch (3.0 eV photons) rings it instantly.

- (a) Define the work function of a metal. [2]
- (b) Explain the two observations using the photon model, including why the red laser's brightness cannot help. [4]
- (c) Calculate the threshold frequency of the photocell. [2]
- (d) State what the wave model of light would (wrongly) predict for the bright red laser, given enough time. [2]

Question 2 — Counting the change (7 marks)

The UV keyring's 3.0 eV photons strike the $\Phi = 2.2 \text{ eV}$ photocell.

- (a) Calculate the maximum kinetic energy of the ejected electrons, in electronvolts and in joules. [3]
- (b) State the stopping potential required to halt even the fastest electrons. [2]
- (c) Doubling the UV intensity doubles something and leaves something unchanged. Identify both. [2]

Question 3 — The electron that thinks it's a wave (8 marks)

Electrons accelerated in a school vacuum tube reach a speed of $4.0 \times 10^6 \text{ m s}^{-1}$ and pass through a graphite film, producing diffraction rings on a screen.

- (a) State the de Broglie hypothesis. [2]
- (b) Calculate the de Broglie wavelength of these electrons. [3]
- (c) Explain why the graphite's atomic spacing ($\sim 2 \times 10^{-10} \text{ m}$) makes it a suitable diffraction grating for them. [1]
- (d) A cricket ball (0.16 kg at 40 m s^{-1}) has a de Broglie wavelength of about 10^{-34} m . Use this number to explain why bowlers need not worry about diffraction. [2]

Answers

Question 1 — The doorbell that refused

- (a) The minimum energy needed to remove an electron from the surface of the metal.
- (b) The key point is that *one photon interacts with one electron*, and its energy cannot be shared or saved up. Red: each photon carries 1.9 eV, less than the 2.2 eV needed, so no single photon can free an electron. Making the laser brighter only delivers *more* photons, each still individually inadequate, so nothing happens however bright it is. Ultraviolet: each photon carries 3.0 eV > 2.2 eV, so every absorbed photon can eject an electron immediately, even from a feeble source.
- (c) $f_0 = \frac{\Phi}{h} = \frac{2.2 \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}} = 5.3 \times 10^{14} \text{ Hz}.$
- (d) The wave model treats energy as arriving continuously and spread over the surface, so an electron should be able to *accumulate* energy gradually. A sufficiently bright beam of *any* frequency would then eject electrons after some delay, and brighter light would give faster electrons. Neither is observed: there is a sharp threshold frequency and no measurable delay.

Question 2 — Counting the change

- (a) $E_{\max} = hf - \Phi = 3.0 - 2.2 = 0.80 \text{ eV}$, which is $0.80 \times 1.6 \times 10^{-19} = 1.3 \times 10^{-19} \text{ J}.$
- (b) $V_s = \frac{E_{\max}}{e} = 0.80 \text{ V}.$ (Numerically equal to $E_{\max}$ in electronvolts, which is the convenience of that unit.)
- (c) *Doubles*: the photocurrent, since twice as many photons per second eject twice as many electrons per second. *Unchanged*: the maximum kinetic energy, and therefore the stopping potential, both of which are fixed by the energy of an individual photon rather than by how many arrive.

Question 3 — The electron that thinks it's a wave

- (a) Every particle has an associated wavelength $\lambda = \frac{h}{p} = \frac{h}{mv}$, so matter shows wave behaviour as well as particle behaviour.
- (b) $\lambda = \frac{6.63 \times 10^{-34}}{(9.11 \times 10^{-31})(4.0 \times 10^6)} = 1.8 \times 10^{-10} \text{ m}.$
- (c) Diffraction is only pronounced when the spacing of the obstacles is comparable to the wavelength. Here the atomic spacing $2 \times 10^{-10} \text{ m}$ and the electron wavelength $1.8 \times 10^{-10} \text{ m}$ are almost identical, so the graphite lattice diffracts these electrons strongly.
- (d) At about 10^{-34} m , the ball's wavelength is roughly 24 orders of magnitude smaller than any gap or obstacle it meets. Diffraction effects scale with the ratio of wavelength to aperture, so that ratio is around 10^{-24} : utterly immeasurable. Classical physics is entirely safe on the pitch.